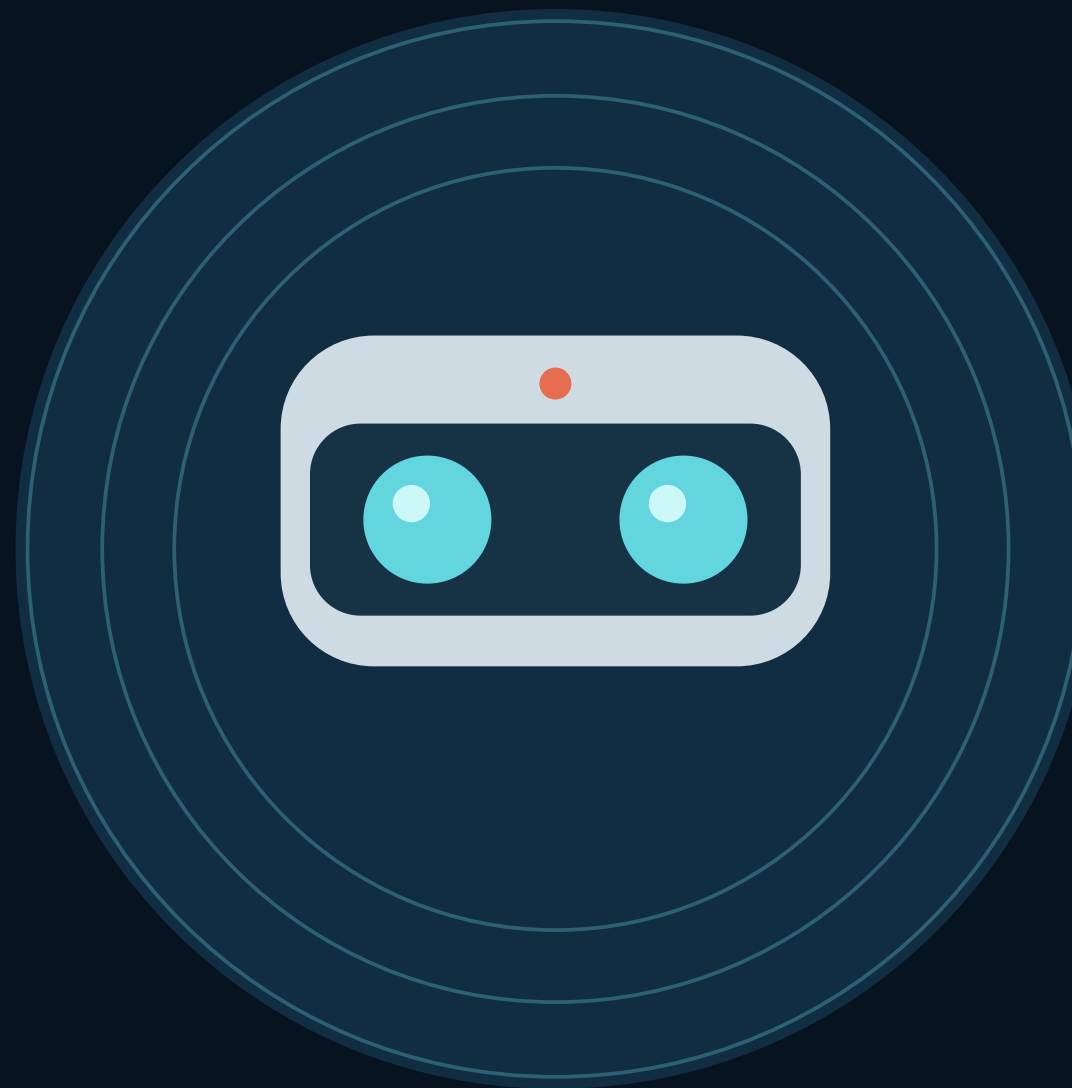


R.O.B. VISION

OPTICAL ROBOT COMPANION / MK.01



MATRIX DISPLAY GUIDE

UNO Q status animations and display states

CURRENT REFERENCE / 24 SEPTEMBER 2026

UNO Q + RETROPIE LIVE · CAMERA VALIDATION PENDING

CHAPTER 01 / CURRENT REFERENCE

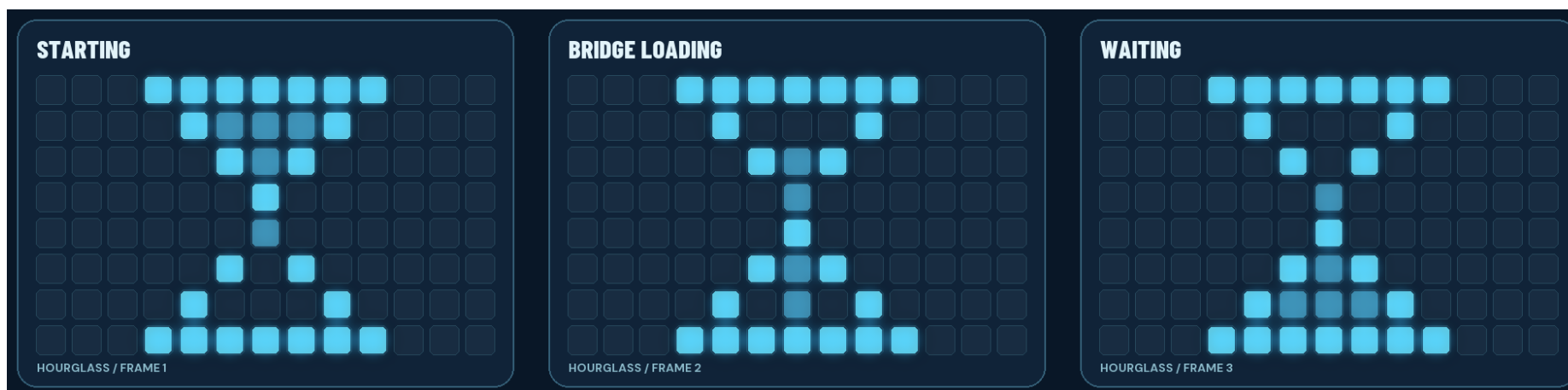
UNO Q matrix display design

Status: implemented in the R.O.B. Vision App Lab sketch and Linux bridge; physical display behavior still needs direct visual confirmation. RetroPie pairing code is generated on RetroPie and entered on the browser Setup page. The UNO Q matrix shows a pairing cue only. This guide defines the behavior for the built-in 13-column by 8-row blue LED matrix. The large browser dashboard remains the place for the full R.O.B. animation, accessories, and explanations.

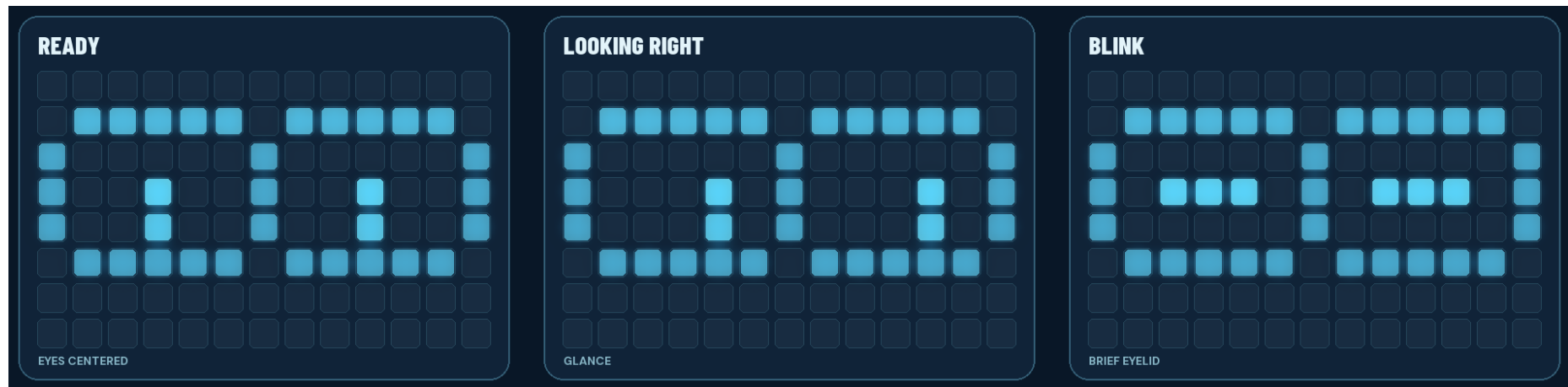
These images show the 13×8 display states defined in `sketch/sketch.ino`. Each panel is one frame of an animation.

The animations

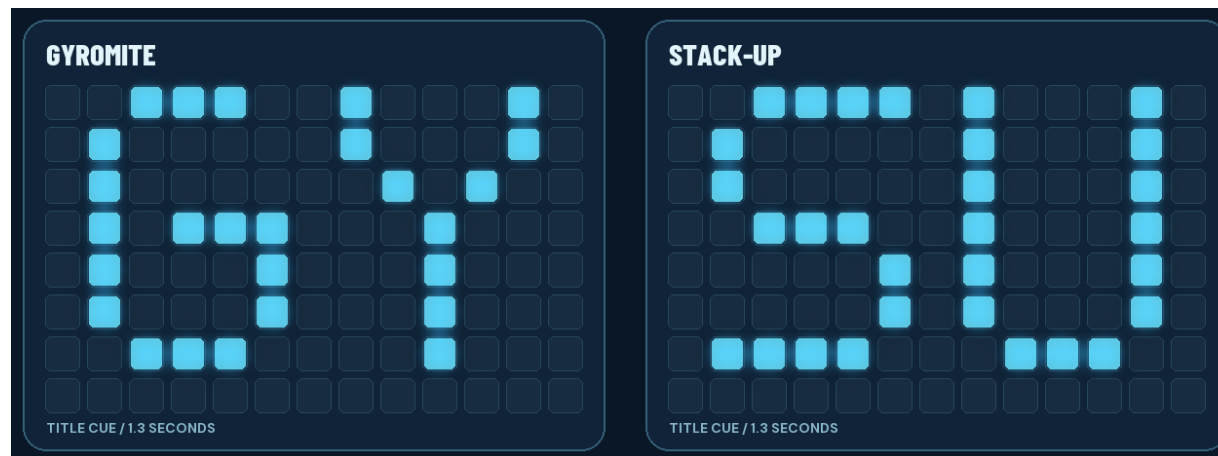
Starting and reconnecting. Three hourglass frames repeat until the controller is ready.

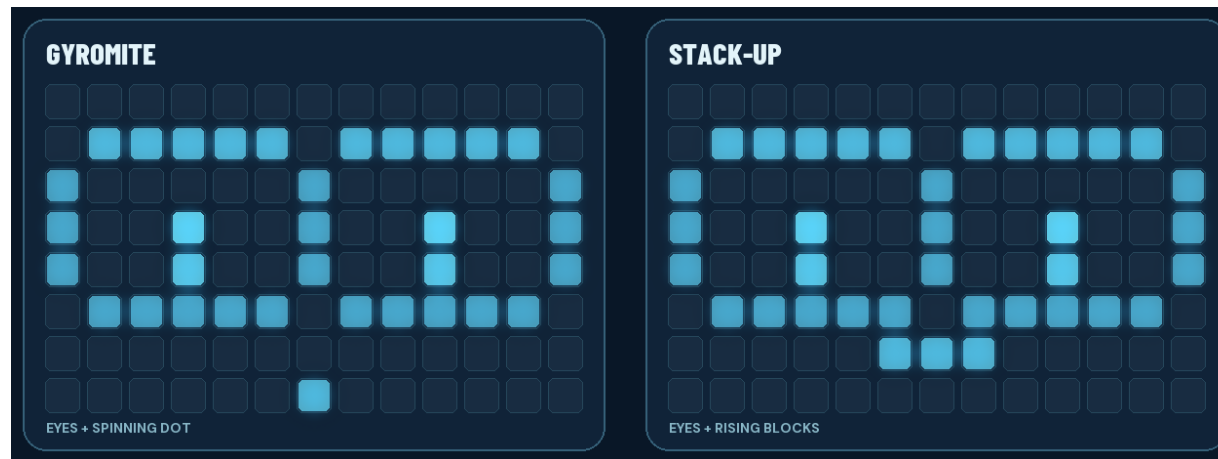


Ready, glance, blink. The eyes move slowly while R.O.B. waits for a game.

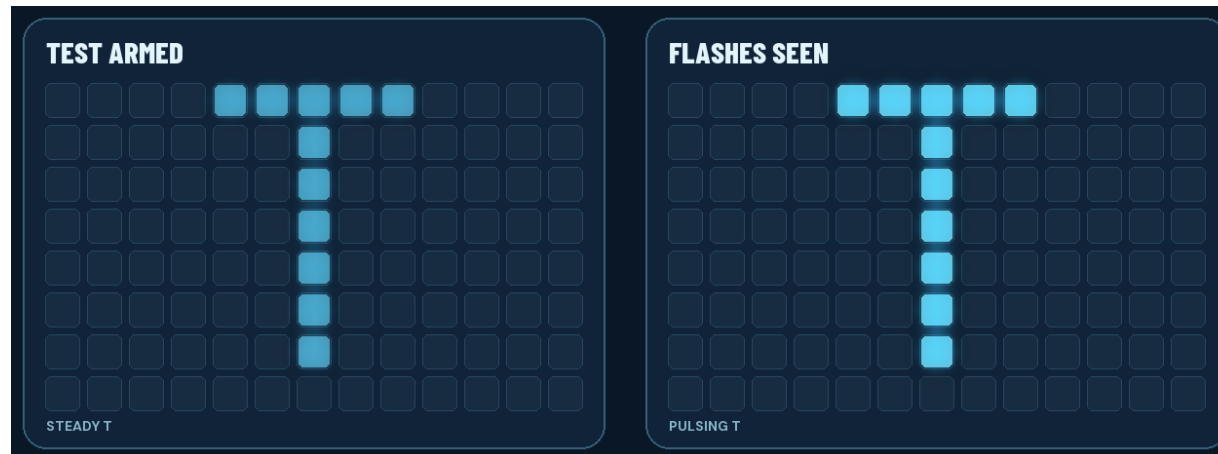


Game selected. A brief title cue identifies the game; the eyes then return with a small decorative accent. The accents do not indicate a physical gyro or block movement.



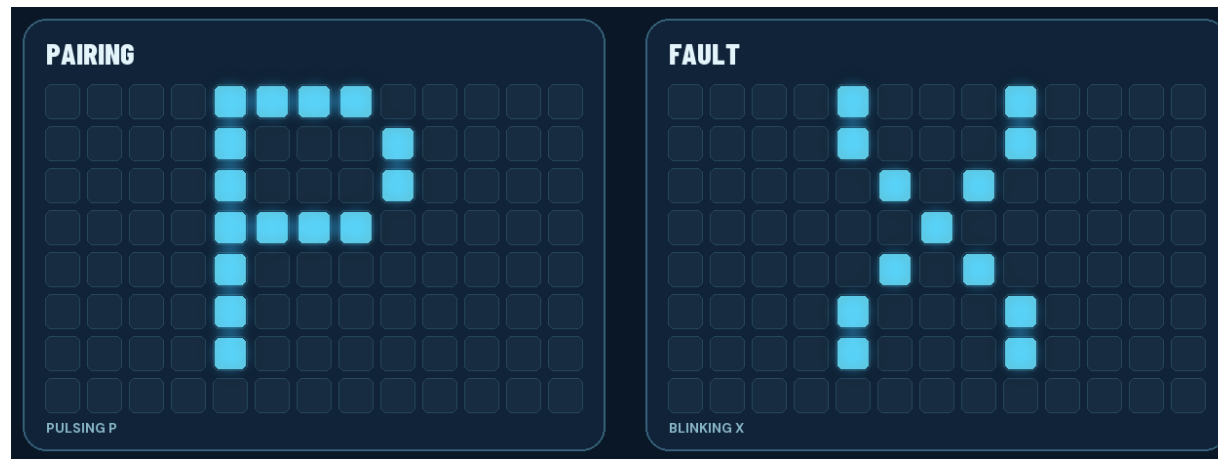


Setup and faults. The T is steady when Test is armed and gently pulses after sustained flashes are detected. P pulses during pairing; X blinks for a fault.



What people should see

State	Matrix	Meaning
Board power-on, before our sketch runs	Arduino's own boot graphics	System startup; R.O.B. Vision cannot own the matrix yet.
R.O.B. Vision sketch starts	VirtualGlove-style pulsing hourglass	The app and Linux side are starting. Begin it as early as the platform permits, before waiting for Router Bridge setup.
Linux bridge connects but the controller is still starting or reconnecting	The same hourglass, restarted or continuing smoothly	The bridge is available; the controller has not yet reported ready. A bridge connection alone must not imply game readiness.
Controller ready, no game selected	A pair of curious R.O.B. eyes: slow left/right glance and occasional blink	R.O.B. is awake and waiting for a game. This is an attract animation, not proof that a camera is attached.
Gyromite selected	Brief GY, then eyes with a small spinning-dot accent	The selected game is Gyromite. The accent is decorative; it must not claim a gyro is actually spinning.
Stack-Up selected	Brief SU, then eyes with a small rising-block accent	The selected game is Stack-Up. The accent is decorative; it must not claim a block was moved.
Valid movement command accepted	Eyes glance in the movement direction, rise/fall for vertical motion, or narrow briefly for grip	Mirror the authoritative virtual action, then return to the selected game's eye loop. Never animate from an undecoded flash alone.
Camera Test check armed	Large T	The user is checking optical alignment. The browser separately indicates whether Test flashes were actually detected.
Test flashes detected	T gently pulses	The camera has observed sustained Test-mode alternation. Avoid rapid full-matrix strobing that could reflect into the camera.
Pairing in progress	Large P with a slow pulse	Open the browser Setup page and read the pairing code and fingerprint shown on RetroPie. Do not show invented or unrelated digits on the UNO Q.
Camera or controller fault	Blinking X	Open Setup for the actual problem; the matrix alone cannot explain it.
App stopped	Matrix released/blank as the platform permits	R.O.B. Vision is no longer driving the display.



The face should be drawn for the physical 13×8 grid, with two distinct eyes and enough dark space to make a blink or sideways glance readable. The blue LEDs are monochrome: eye shapes and brightness, not color, carry the expression. Limit idle motion to an occasional glance or blink so it feels alive without competing with the game or affecting the camera. Show GY/SU long enough to identify the title, then let the eyes take over. A brief repeat of the title mark after a long idle period is acceptable.

State ownership and priority

The sketch owns all framebuffer writes and animates the startup hourglass without waiting for Linux. Linux sends compact requested states and accepted action hints through Router Bridge, using the UNO Q controller's authoritative game, camera, test, and fault state. It sends a heartbeat at least once per second. If Linux disappears for about 3.5 seconds, the display returns to the hourglass rather than freezing a misleading game face. The sketch cannot replace Arduino's protected boot logo before App Lab releases the microcontroller.

Display priority: **fault** → **pairing** → **Test check** → **startup/reconnect** → **selected game** → **idle eyes**. A movement hint is temporary within the selected game and never overrides fault, pairing, or Test. A camera being absent should be visible on Setup; the idle eyes alone must not claim optical readiness. Pairing has a bounded lifetime and returns to the prior state when complete or expired.

Pairing distinction

VirtualGlove has a separate physical pairing display for its own device approval PIN. R.O.B. Vision pairs **RetroPie to the UNO Q**: RetroPie produces a six-digit code and certificate fingerprint, while the UNO Q browser Setup page accepts them. The matrix P is a status cue when **Pair another console** is opened or pairing is submitted; it expires or clears afterward. Showing the code on the UNO Q would require a deliberate change to the pairing protocol.

Physical acceptance checks still needed

1. Cold boot shows the protected system graphics, then the hourglass as soon as the R.O.B. Vision sketch can run; the hourglass remains through Router Bridge initialization.
2. When the Linux controller is ready, the hourglass changes to the idle eyes. Stopping/restarting Linux restores the hourglass and then the correct current game display.
3. Real Gyromite and Stack-Up launch/exit notifications select and clear their distinct marks and eye loops.
4. Test T, pairing P, and fault X follow the priority and timeout rules. No matrix state claims a camera signal or game action that the controller has not observed.
5. Verify brightness and animation beside the intended camera/display setup to ensure matrix reflections do not trigger optical recognition.